Radical Pair Hamiltonian Parameter Comparison

Convention: all Hamiltonian coefficients are reported as frequencies, A/h, J/h, and |D|/h in MHz. The point-dipole prefactor is |D|/h = 52.04 MHz nm^3 / r^3. Some OOP-ESEEM papers report a different dipolar D convention, approximately D_OOP = -3 D / 2; this table uses |D|/h to avoid ambiguity.

System	Radical pair	r (A)	J/h (MHz)	|D|/h (MHz)	Dominant A/h (MHz)	Notes / provenance
Cryptochrome	FAD radical anion / TrpD radical cation	21.0 ErCry4; 21.2 +/- 0.2 pigeon Cry4	~0.03	5.6; pigeon 5.46 +/- 0.15	FAD N5 principal: -2.2, -2.0, 47.5. Trp Hbeta1: 42.0, 43.7, 49.2. Trp N1 max: 30.3.	Distance and J from OOP-ESEEM literature. Hyperfine values from flavin/Trp radical datasets. Strong A terms are comparable to or larger than D.
Photolyase	Measured photoreduction pair: FAD radical anion / TrpC radical cation. Repair pair: FADH radical / CPD radical anion.	19.5 +/- 0.5 for E. coli photolyase FAD-TrpC	~0.05 for FAD-TrpC	7.02 +/- 0.54	For measured FAD-TrpC pair, use FAD/Trp values above. CPD repair-pair A values still need a dedicated source.	Recommend keeping this distinction explicit: FAD-TrpC is well measured; FADH/CPD is the direct repair chemistry but less cleanly parameterized here.
Photosystem I / photosynthetic RC	P700 radical cation / A1 radical anion	25.4 +/- 0.03	Likely << |D|; often unresolved/small for long-lived separated pair	3.18 +/- 0.01	P700+ donor: 1H CH3 (Aperp,A||) = (5.1,6.4), (3.1,4.3), (2.5,3.7) MHz; 15N A||/Aiso = 5.85/3.51, 5.00/2.90, 2.80/1.68 MHz. A1-: nearby 15N Riso/T = 1.98/0.14 and 0.33/0.58 MHz.	Measured/fitted ENDOR/ESEEM. P700+ 15N errors: A|| +/-0.2 MHz and Aperp +/-0.1 MHz for main tensors; ENDOR +/-0.05 MHz. A1- ESEEM frequency error +/-0.125 MHz.
Bacterial reaction center	P865 radical cation / QA radical anion	28.4	Likely small for separated pair	2.27	P865+ donor: 1H beta-L = 9.70, 8.70, 6.25, 4.50 MHz; methyl = 5.70, 3.96, 3.20, 1.37 MHz; 15N A||/Aiso = 6.42/3.72, 5.78/3.49, 5.22/2.97, 4.67/2.61 MHz. QA- model: 17O O1 (-95,19,19), O4 (-80,13,14); 13C C4 (41,-9,-11), C1 (-20,-16,-11) MHz.	P865+ values measured/fitted by ENDOR/ESEEM. 1H errors: methyl +/-0.03 MHz, beta +/-0.05 MHz; 15N: Aperp +/-0.1, A|| +/-0.2 MHz. QA- values are DFT model-B total principal tensors for the Rb. sphaeroides QA site, benchmarked against ENDOR data.
Carotenoid-porphyrin-fullerene triad	Carotenoid radical cation / fullerene radical anion	TBD centroid distance	Neglected in published CPF spin-dynamics models; assumed small	Neglected in published CPF models; structure-derived point-dipole estimate pending	Carotenoid 1H isotropic A/h: largest +16.23, -10.81, -10.12, +9.22 x3, -8.88, +8.55 x3 MHz; 45 proton couplings, RMS 39.1 MHz. C60 treated hyperfine-free.	Lindoy/Lewis CPF model uses g = ge, ignores 13C, and uses kS = 1.8e7 s^-1, kT = 7.1e4 s^-1. Scalar exchange and anisotropic dipolar coupling are explicitly neglected.
AsLOV2 WT	Triplet FMN / C450 contact intermediate leading to cysteinyl-flavin adduct	Likely <10 A contact scale	Strong/contact; not cleanly measured as weakly coupled RP	>52 if r < 1 nm; ~416 at r = 0.5 nm	Flavin neutral examples: H5 -21.6; N5 11.0; N10 5.9.	Strong-coupling comparator. The row should be framed as contact flavin-cysteine photochemistry, not a measured separated RP Hamiltonian.
MagLOV	Likely FMN radical anion / protein donor radical cation; later FMNH radical chemistry	Donor not pinned	Not measured	Not measured	If Trp donor: Hbeta1 ~45. If flavin anion: N5 principal max 47.5.	Excellent MFE/ODMR sensor row, but microscopic J, D, and donor geometry are not yet measured in the same way.

Source Notes

- OOP-ESEEM distances/J discussion: Al Said, Weber, and Schleicher, Front. Mol. Biosci. 2022, doi: 10.3389/fmolb.2022.890826.
- RadicalPy hyperfine database and supplement: Evans et al., J. Chem. Theory Comput. 2024, doi: 10.1021/acs.jctc.4c00887.
- Chemical compass triad benchmark: Maeda et al., Nature 2008, doi: 10.1038/nature06834.
- MagLOV source: Abrahams et al., bioRxiv 2025, doi: 10.1101/2024.11.25.625143.
- CPF hyperfine/model parameters: Lewis et al., J. Chem. Phys. 2014, doi: 10.1063/1.4890659; Lewis et al., J. Chem. Phys. 2018, doi: 10.1063/1.5038558; Lindoy et al., J. Chem. Phys. 2020, doi: 10.1063/5.0006411.
- Photosystem I hyperfines: Kaess et al., Chemical Physics 1995, doi: 10.1016/0301-0104(95)00021-F; Kaess and Lubitz, Chem. Phys. Lett. 1996, doi: 10.1016/0009-2614(96)00129-7; ESEEM study of A1 phyllosemiquinone, Biochemistry 1997, doi: 10.1021/bi971360a.
- Bacterial RC hyperfines: Kass et al., J. Phys. Chem. 1995, doi: 10.1021/j100001a065; Rautter et al., Biochemistry 1997, doi: 10.1021/bi962859s; Lendzian et al., Photosynth. Res. 1998, doi: 10.1023/A:1006030221445.
- QA- ubisemiquinone hyperfine source: O'Malley, Chem. Phys. Lett. 1998, doi: 10.1016/S0009-2614(97)01456-5; reports B3LYP/EPR-II model-B QA-site 17O and 13C total principal hyperfine tensors in MHz.